MILITARY SPECIFICATION

STEEL BARS, WIRE, FORGING STOCK, FORGINGS, MECHANICAL TUBING, (431) PREMIUM QUALITY

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

1.1 This specification presents requirements for a premium quality of type 431 corrosion-resistant steel.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-H-6875	Heat Treatment of Steels (Aircraft Practice, Process for)

STANDARDS

Federal

FED. STD. NO. 48	Tolerances for Steel and Iron Wrought Products
FED. TEST METHOD STD. NO. 151	Metals; Test Methods
FED. STD. NO. 183	Continuous Identification Marking of Iron and Steel Products

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-753	Corrosion - Resistant Steel Parts, Sampling, Inspection and Testing for Surface Passivation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9510

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein:

Aerospace Materials Specification

AMS2300 Premium Aircraft Quality Steel Cleanliness, Magnetic Particle Inspection Procedure

(Copies of AMS specifications may be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Ave., New York, N. Y. 10017.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, Ill., 60606.)

3. REQUIREMENTS

3.1 Chemical composition.- The chemical composition shall be as specified in table I.

TABLE I. Chemical composition

Element	Percent	Check analysis tolerance (percent) 1/
Carbon	0.13 - 0.17	0.01
Manganese	0.30 - 0.80	.03
Silicon	0.20 - 0.60	.05
Phosphorous	0.025 (max.)	.005
Sulfur	0.025 (max.)	.005
Chromium	15.50 -16.50	.20
Nickel	2.00 - 3.00	.07
Molybdenum	0.25 (max.)	.03
Nitrogen	0.10 (max.)	.01
Iron	Balance	-

1/ Individual determinations may vary above or below the specified range to the extent shown in the check analysis column, except that elements in any heat shall not vary both above and below the specified range.

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3.2 Quality.- All material supplied to this specification shall be produced by the consumable electrode remelt process, or other vacuum melting process approved by the procuring agency, and magnetic particle inspected in accordance with the requirements of AMS2300, except that limits upon frequency and severity rating shall be as agreed upon by purchaser and supplier. However, the steel producer shall submit, for each heat produced to this specification, the average and highest individual ratings to the Naval Air Systems Command, Attn: AIR-52031, Washington, D. C. 20360.

3.2.1 Microstructure.- After heat treatment to the HT-125 condition, materials shall not exhibit banding or segregation in excess of the limits of 3.2.1.1 or 3.2.1.2, respectively. The limits apply to all fields of longitudinal sections of specimens when examined at 250 diameters magnification.

3.2.1.1 Free ferrite.- Material shall not contain banded free ferrite in excess of that indicated by figure 3. Blocky ferrite (not banded) is acceptable, provided the aggregate area does not exceed 4 percent.

3.2.1.2 Retained austenite.- Material shall not contain banded retained austenite in excess of than shown on figure 5. Figures 6 and 7 represent unacceptable microstructures. Blocky austenite not banded is permitted.

3.3 Condition as supplied.- The material shall be furnished annealed, or otherwise heat treated to a hardness range of Brinell 229-302 (excluding forging stock).

3.3.1 Surface condition.-

3.3.1.1 Bar and forging stock, round.- Surfaces shall be ground or rough turned.

3.3.1.2 Bar and forging stock, rectangular.- Surfaces shall be hot-rolled and free from scale. Slight surface imperfections which are removable within half the specified tolerances will not be cause for rejection.

3.3.1.3 Tubing.- Surface imperfections such as handling marks, straightening marks, light mandrel and die or roll marks, shallow pits, and scale pattern will not be cause for rejection, provided the imperfections are removable without reducing the diameter or wall thickness of the tubing below the permissible tolerance limits.

3.3.1.4 Wire.- Surfaces shall be cold drawn, ground, polished or pickled.

3.3.2 Grain size.- Grain size shall be No. 3 or finer, as indicated in Supplement A of Federal Test Method Standard No. 151, when tested in the HT-125 condition (excluding reforging stock).

3.4 Response to thermal treatment.- When heat treated as specified herein, the material shall develop the properties specified in table II (see 3.4.1 and 3.4.2).

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3.4.1 HT-200 condition.- HT-200 is achieved by (1) quenching in oil at 52° ±6° C (125° ±10° F) from 1024° ±14° C (1875° ±25° F); (2) cooling to room temperature in water; (3) soaking at -73° ±11° C (-100° ±20° F) in a suitable refrigerant for 2 hours; (4) tempering at 288° ±14° C (550° ±25° F) for 2 hours; and (5) retempering at 288° ±14° C (550° ±25° F) for 2 hours.

3.4.2 HT-125 condition.- HT-125 condition is achieved by quenching in agitated oil from a temperature of 1875° ±25° F (1024 ±14° C), cooling to room temperature in water, and tempering at 1175 ±25° F (635° ±14° C) for 3 hours. The heat treatment is used to provide the necessary contrast for microstructure evaluation.

TABLE II. Mechanical properties

Direction of testing and condition 1/, 2/	Ultimate tensile strength (minimum psi)	Yield strength at 0.2% offset or at extension indicated (E 28,000,000)		Elongation in 4 diameters (minimum percent)	Reduction of area (minimum percent)	Charpy impact (minimum ft. lbs.)
		Minimum psi	2 inch gage length (inch)			
Longitudinal HT-200	200,000	150,000	0.0150	10	40	35
Transverse HT-200	200,000	150,000	0.0150	5	27	-

1/ Specimens shall have a minimum diameter of 0.250 inch. When final size is to small to accommodate the specimens, specimens may be selected at a prior fabrication stage.

2/ Specimens shall be taken in the short transverse direction across the central axis for the transverse.

3.5 Dimensions and tolerances.-

3.5.1 Diameter or thickness.- Variations of measured dimensions from ordered dimensions shall be within the tolerance limits specified in Federal Standard No. 48.

3.5.2 Exact lengths.- Bars of all sizes may be ordered to exact lengths or in lengths expressed as a multiple of a definite unit, with tolerances as specified in table titled, "Stainless and Heat Resisting Steels, Hot Rolled Bars, Length Tolerances", of Federal Standard No. 48.

3.5.3 Mill lengths.- When exact or multiple lengths are not ordered, bars will be accepted in mill lengths of 8 to 20 feet, but not more than 10 percent of any order shall be furnished in lengths shorter than 10 feet.

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3.6 Identification of product.- Bars shall be marked in accordance with Federal Standard No. 183. The following marking items shall be included in the legend:

MIL-S-8967
Heat number
Manufacturer's name or identifying trade mark.

3.6.1 Bars and wire smaller than 1/2 inch in diameter or 3/8 inch in width of flat shall be tagged with the information above at each bundle end and with an extra tag included in the bundle.

3.7 Workmanship.- Material shall be sound and of uniform condition, smooth, clean, and free from scale and injurious defects, such as cracks, seams, tears, grooves, laminations, pits, and blisters. Slight surface imperfections which are removable within half the specified tolerances will not be considered injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- All the tests required herein for the testing of steel are classified as quality conformance tests, for which necessary sampling techniques and methods of testing are specified in this section.

4.3 Sampling and inspection for dimensions, finish, workmanship, packaging, and identification-marking requirements.- Units of product shall be randomly selected to represent each lot of material of one heat, nominal dimensions, and offered for acceptance at one time, in accordance with table III.

4.3.1 Method.- Samples shall be visually inspected.

4.4 Chemical analysis.-

4.4.1 Sampling.- Samples for check chemical analysis shall be selected in accordance with Method 111 of Federal Test Method Standard No. 151, to represent each heat in the shipment.

4.4.1.1 Sample location.- Samples for check chemical analysis shall be taken parallel to the axis of the material selected at a point midway between the center and surface, except that material less than 1-1/4 inches thick shall be sampled through the entire cross section. The sample shall consist of not less than 2 ounces.

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TABLE III. Sampling for examination of product

Lot size	Sample size	Acceptance number
1 to 7	All	0
8 to 40	7	0
41 to 110	15	1
111 to 180	25	2
181 to 301	35	3
Over 301	50	4

4.4.1.2 Waiver.- Samples for check chemical analysis may be waived provided that all of the material under inspection can be identified as being made from a heat previously analyzed and found to be in conformance with the chemical composition specified herein.

4.4.1.3 Additional samples.- Where the material is not identifiable as to heat or where the identity of any portion of the shipment is obscure in any respect, samples shall be selected in accordance with table III.

4.4.2 Method.- Analysis shall be in accordance with Methods 111 or 112 of Federal Test Method Standard No. 151, except that in the event of dispute, Method 111 shall be used.

4.5 Microstructure and grain size.-

4.5.1 Sampling.- A sample shall be selected to represent each 50 pieces of each size from each heat except when the minimum cross sectional dimension exceeds 2 inches a sample shall be selected to represent each 10 pieces. When materials are not identified or the identity is obscure in any respect, sampling shall be in accordance with table III. The steel manufacturer may use heat evaluation tests in lieu of testing finished mill products (see 4.4.1.3).

4.5.2 Method.- Specimens shall be heat treated to the HT-125 condition, sectioned parallel to the direction of rolling or drawing, to expose surfaces as indicated by figure 8, shall be suitably polished, etched with Villella's etchant, and examined at 250 diameters magnification. Acceptance standards are as indicated by 3.2.1 and figures 1 through 7.

4.6 Response to heat treatment.-

4.6.1 Sampling.- One tension test sample shall be selected from each 500 or less pieces of mill product of each configuration and size, produced under the same processing conditions, from the same heat, and submitted for acceptance at one time. When materials are not identified or the identity is obscure in any respect, sampling shall be in accordance with table III. The steel manufacturer may use heat evaluation tests in lieu of finish testing.

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FIGURE 1. Desired microstructure (100% Martensite)

Etchant-Villella's
MAG: 250X



FIGURE 2. Acceptable microstructure containing small amount of banded free ferrite

Etchant-Villella's
MAG: 250X

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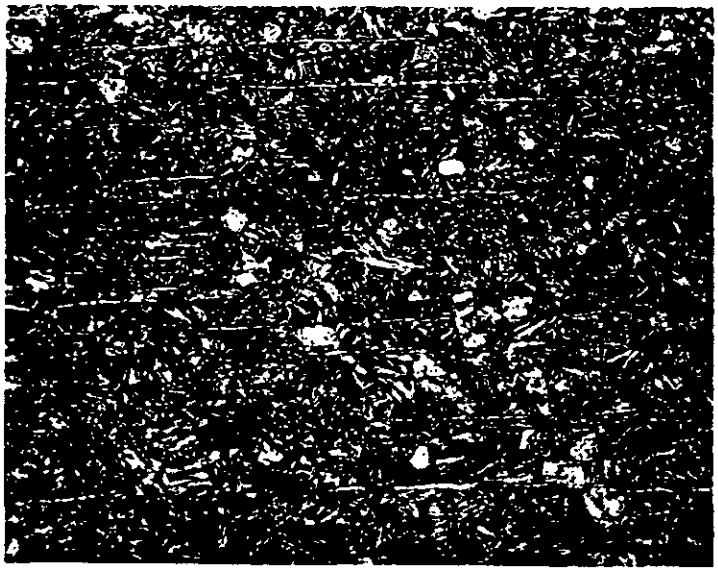


FIGURE 3. Acceptable microstructure containing maximum amount of banded free ferrite permitted in any field

Etchant-Villella's
MAG: 250X

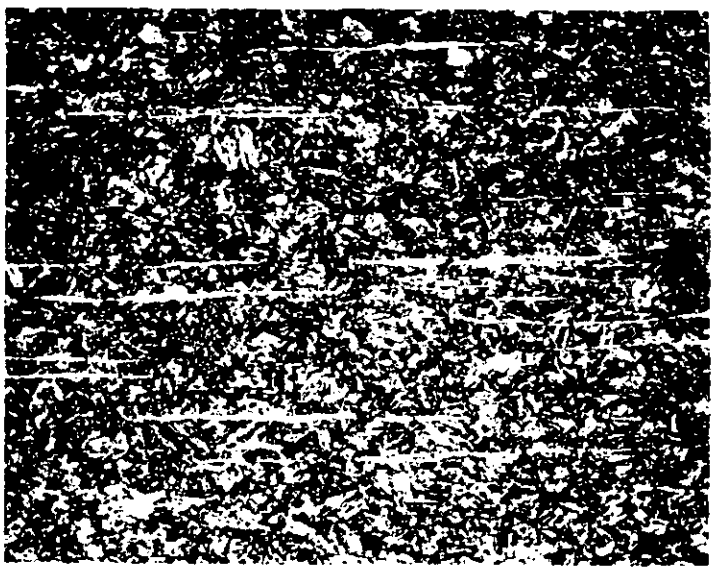


FIGURE 4. Unacceptable microstructure containing excessive banded free ferrite

Etchant-Villella's
MAG: 250X

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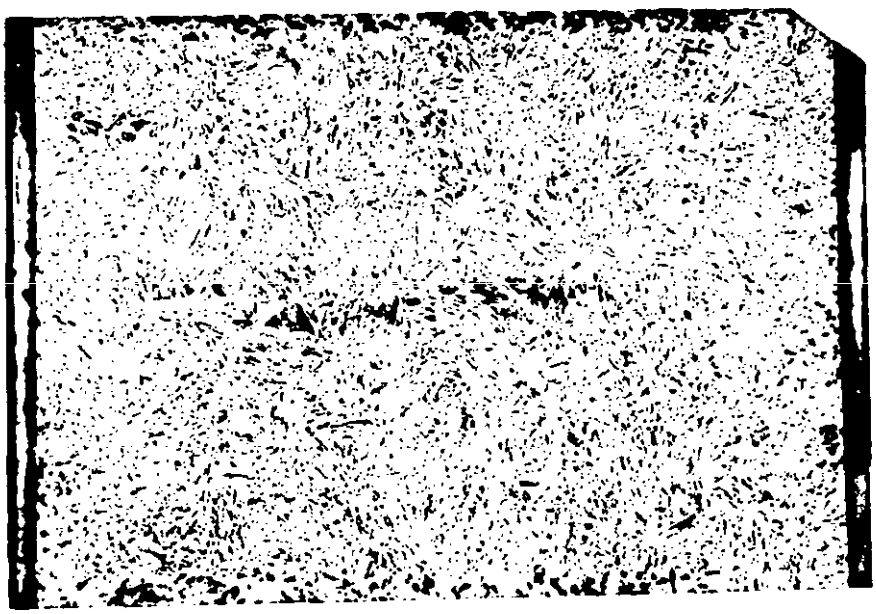


FIGURE 5. Acceptable microstructure containing maximum amount of banded retained austenite permitted in any field of examination

Etchant-Villella's
MAG: 250X

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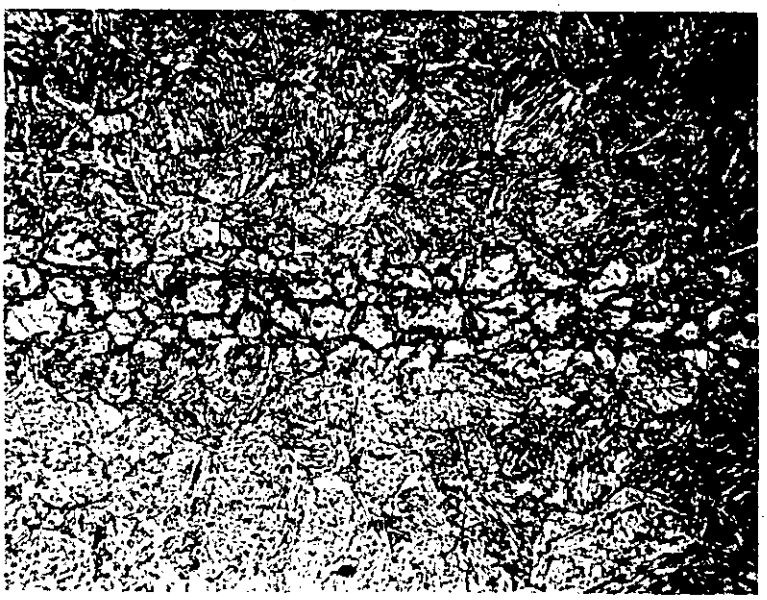


FIGURE 6. Unacceptable microstructure containing excessive banded retained austenite

Etchant-Villella's
MAG: 250X

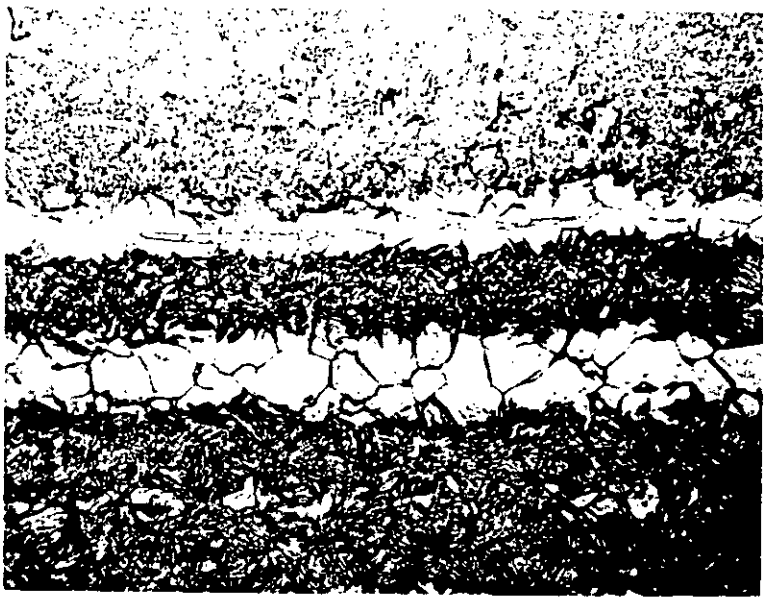


FIGURE 7. Unacceptable microstructure containing excessive banded retained austenite

Etchant-Villella's
MAG: 250X

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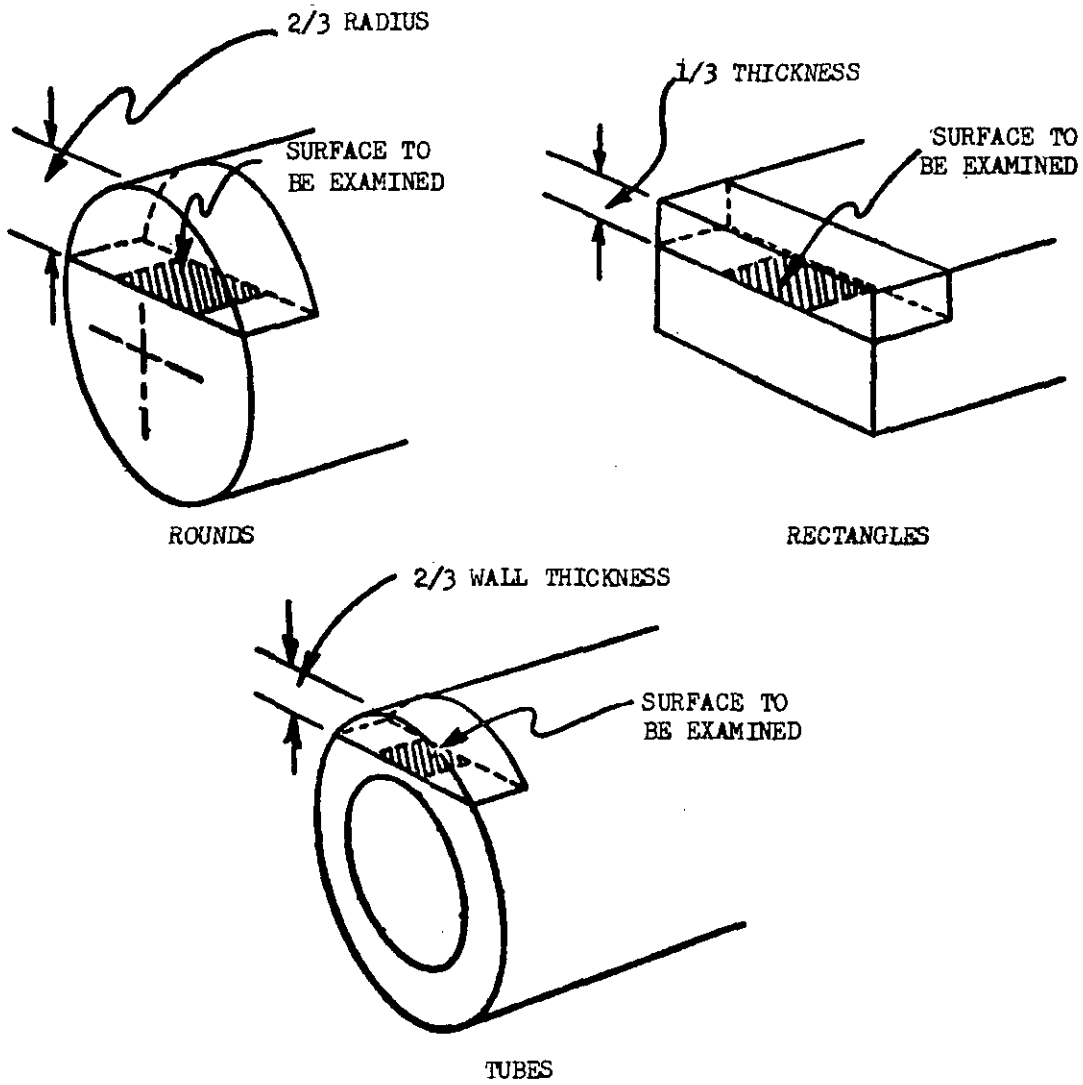


FIGURE 8. Location of micro-specimens

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4.6.1.1 Heat lots.- Heat lots may be certified by tests of four samples from intermediate mill products of sufficient size to provide short of center transverse specimens for testing.

4.6.2 Preparation of specimens.- All tension and Charpy impact test samples shall be machined to dimensions before heat treatment, except for finish machining of bearing and close-tolerance surfaces. Tension specimens of Method 211 of Fed. Test Method Std. No. 151 and Charpy impact specimens of Method 221 of Fed. Test Method Std. No. 151 shall be used.

4.6.2.1 Specimens shall be heat treated in accordance with equipment and process requirements of MIL-H-6875.

4.6.3 Method.- Tension tests and determination of yield strength shall be conducted in accordance with Method 211, and Charpy impact tests shall be conducted in accordance with Method 221 of Fed. Test Method Std. No. 151.

4.7 Rejection.- When the failure of one or more specimens indicate that the test sample fails to meet a specified requirement, the entire lot shall be rejected.

4.7.1 Retest.- At the discretion of the supplier, retest will be permitted. A retest sample of five specimens shall be tested to replace each failed specimen of the initial sample. If one or more of the retest specimens fail, the lot shall be rejected with no further testing permitted.

4.7.2 Resubmittal.- If the defectives in a lot previously rejected can be identified and reprocessed or removed, resampling at frequency of the initial sampling will be acceptable.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.-

5.1.1 Level A.- Material shall be prepared for shipment in accordance with MIL-STD-163.

5.1.2 Level C.- Material shall be packed in boxes or crates so constructed as to insure acceptance by common or other carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules or regulations of other carriers, as applicable to the mode of transportation.

5.2 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and MIL-STD-163. The identification shall include the following information listed in the order shown:

(a) Stock No. or other identification number as specified in the purchase document

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(b) Steel (bars, wire, forging stock, forgings, or tubing, as appropriate), 431, premium quality MIL-S-8967, size and shape (round, square, hexagon, or flat)
(c) Heat No.
(d) Quantity contained, feet or pounds (as defined in the contractor's order for each size and shape)
(e) Name of contractor (and name of manufacturer, if not the same).
(f) Contract or order No.

6. NOTES

6.1 Intended use.- The steel covered by this specification is intended for use in the manufacture of highly stressed structural parts exposed to corrosive environments and operating temperatures to 500° F for the HT-200 condition. The material is useful for one component of threaded stainless steel fasteners because of a tendency to resist seizure when in contact with other types of corrosion resisting steel.

6.2 Precautions.-

6.2.1 Heat treatment.- This material is not recommended for use after heat treatment to a strength level other than specified in table II because of the probability of developing low impact strength and unsatisfactory corrosion resistance.

6.3 Passivation.- The surfaces of parts manufactured from this type of steel will be passive when tested in accordance with MIL-STD-753.

6.4 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity, thickness, size or shape, (see 3.3).
(c) Level of packaging and packing desired.

6.5 Cold heading wire is not covered by this specification and should be purchased to ASTM A-493.

6.6 Definitions.-

6.6.1 Wire.- Cold finished only: round, square, octagon, hexagon and shape wire, 1/2 inch and under in diameter or size. Cold finished only: flat wire, 1/16 inch to under 3/8 inch in width, 0.010 to under 3/16 inch in thickness.

Custodians:
Navy - AS
Air Force - 11

Preparing activity:
Navy - AS

Project No. 9510-N002

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POSTAGE AND FEES PAID
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OFFICIAL BUSINESS

Aeronautical Standards Group (ASG)

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Silver Spring, Md. 20910

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-S-8967(ASG) Steel Bars, Wire, Forging Stock, etc., (431), Premium Qual.

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?
   ( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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